

Subject Code: ES 102

Environmental Studies: Scientific and Engineering Aspects

ENVIRONMENTAL POLICY AND LEGISLATION

Unit – 6
(Lecture 17-19)

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Importance of Translation in Environmental Law





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What will you learn in Today's Lecture

- Various aspects associated with Environmental Regulations and their importance
- Process of framing an Environmental Policy
- Environmental Policies and Regulations in the Indian Context

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What is Environmental Policy?

Environmental policy encompasses an organization's or government's commitment to laws, regulations, and mechanisms governing environmental issues.

Key Components

- Legislative frameworks
- Regulatory standards
- Enforcement mechanisms
- Monitoring systems
- Compliance measures

Primary Objectives

- Protect natural resources
- Prevent pollution
- Ensure sustainability
- Safeguard public health
- Promote conservation

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What is an Environmental Policy?

- **Policy** refers to a set of principles or plans agreed upon by a government or an organization to be carried out in a particular situation. It refers to the general set of plans and principles for addressing problems and guiding decision-making.
- **Public policies are laws, regulations, orders, incentives, and practices laid by the government with the intention to advance societal welfare.**
- **Environmental policy** is defined as any action deliberately taken to manage human activities with a view to preventing, reducing, or mitigating harmful effects on nature and natural resources, and to ensure that man-made changes to the environment do not have harmful effects on humans or the environment.
- Environmental policy usually covers air and water pollution, waste management, ecosystem management, biodiversity protection, and the protection of natural resources, wildlife, and endangered species. Proper policies and legislation at the national and international levels can reduce venomous pollution and help protect biodiversity and natural resources.
- Environmental policy is probably the only policy domain that has the main goal of protecting non-human entities.



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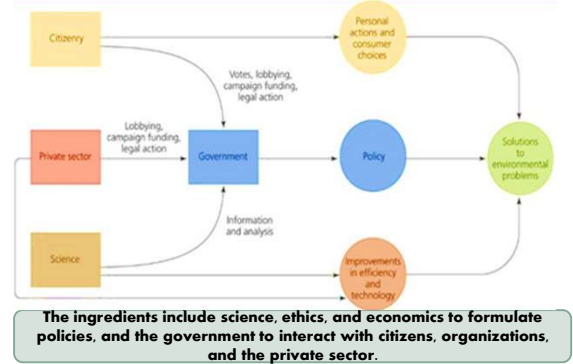
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Environmental Policy

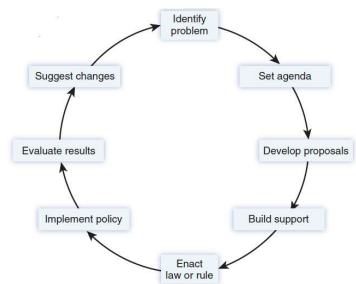
- ❑ It is **primarily concerned with how to govern the relationship between humans and the natural environment in a mutually beneficial manner.** Traditionally, it has been defined in terms of the problems it addressed, such as controlling pollution and waste flows and limiting habitat loss.
- ❑ However, policymakers have begun to appreciate that environmental policy will only be genuinely successful when it is sensitively integrated with other sectors and policy areas.
- ❑ Politically speaking, environmental policy integration represents a new but potentially difficult stage in the continuing metamorphosis of environmental policy into a much wider policy supporting sustainable human development.
- ❑ Environmental policies target sustainable exhibitions, sustainable procurement of resources, sustainable design, prefer going green on energy resources, pollution reduction, conservation of natural resources, and create sustainable awareness

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The policy cycle is a path through which rules are developed, enacted, tested, and revised.



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Historical Development of Environmental Laws

1960s-1970s

Environmental Movement Era

Rise of environmental consciousness; establishment of EPA (1970); Clean Air Act (1970); Clean Water Act (1972)

1980s

Regulatory Expansion

Comprehensive Environmental Response Act (CERCLA/Superfund); Montreal Protocol (1987)

1990s

Global Cooperation

Rio Earth Summit (1992); Kyoto Protocol (1997); sustainable development frameworks

2000s-Present

Climate Focus

Paris Agreement (2015); circular economy policies; net-zero commitments

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Key Environmental Milestones

1962

Silent Spring

Rachel Carson's book sparked environmental movement

1970

First Earth Day

20 million Americans participated

1970

EPA Created

US Environmental Protection Agency established

1987

Brundtland Report

Defined sustainable development

1992

Rio Summit

Agenda 21 and biodiversity convention

2015

Paris Agreement

Global climate action framework

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Need for Environmental Policy & Legislation



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Objectives of Environmental Policy

This originated with the US National Environmental Policy Act (NEPA) on Jan 1st, 1970. A policy is being framed that will encourage productive and enjoyable harmony between man and his environment. The objectives include:

- To promote efforts that will prevent or eliminate damage to the environment and biosphere and stimulate the health and welfare of man.
- To enrich the understanding of the ecological system and natural resources important to the nation.
- Protect natural resources such as water, air, and soil.
- Protection of biodiversity from destruction.
- Prevent pollution.
- Combating desertification and saving landscapes.
- Encourage, develop, and apply the best available technical solution.
- Ensure that the polluters pay for the pollution and control arrangements.
- Focus protection on the heavily polluted areas and river stretches.
- The integrated management of waste and hazardous materials and chemicals.
- Involve the public in decision-making.



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Making an Environmental Policy



1. Identify a problem



3. Envision solution



2. Pinpoint the cause



4. Get organized

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Making an Environmental Policy



5. Cultivate access and influence



6. Shepherd the solution to law



7. Implement, assess and interpret policy

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What is Environmental Legislation?

- **Environmental legislation** is a set of laws and regulations which aim at protecting the environment (like protecting the quality of air, water, wildlife strength, and environmental health, etc) from harmful actions. It means a plan of action adopted by the Government rationalizing the course of action.
- The **Environmental legislation** tries to enact all the laws of the environment in a justifiable manner
- Legislation may take many forms, including regulation of emissions that may lead to environmental pollution, taxation of the environment- and health-damaging activities, and establishing the legal framework for trading schemes, for example, carbon emissions. Other actions may rely on voluntary agreements. Among major current legislative frameworks are those relating to environmental permitting, and those mandating environment and health impact assessments.
- The Legislation varies from country to country, but their main focus is on a common goal of secure and sustainable living on Earth.



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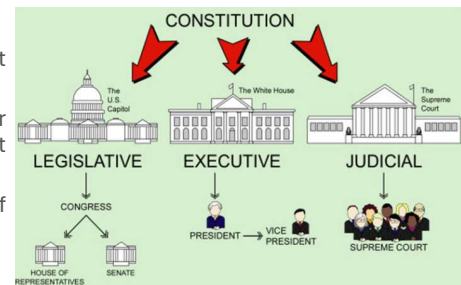
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Environmental Legislation

The umbrella of environmental legislation encompasses many laws and regulations, yet they all work together toward a common goal: regulating the interaction between humans and the natural world to reduce threats to the environment and improve public health.

To better understand environmental law, let's look at an example.

- ❑ An energy company wants to build a coal-burning power plant to create electricity for the community.
 - ❑ Where should this power plant be built?
 - ❑ What type of pollutants might result from the coal burning, and what measures will need to be taken to control harmful emissions?
 - ❑ If the power plant is built at the edge of town to lessen air pollution for the human population, how will this impact lesser species that inhabit the land downwind of the plant?
 - ❑ These are all considerations to be evaluated within the scope of environmental law.



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Environmental Legislation serves the following purposes

- ❑ It protects the environment by punishing offenders
- ❑ It protects the resources from abuse
- ❑ Encourage the restoration of the environment
- ❑ It restricts the emission or discharge of environmental pollutants in excess of the standards.
- ❑ It ensures a smooth solution of conflicts.
- ❑ It serves as guidelines for industries:
 - ❑ To select non-objectionable raw materials for their processes.
 - ❑ To make improvements in manufacturing processes.

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India's Environmental Legislation

Constitutional Provisions

Article 48A: State shall protect and improve environment and safeguard forests and wildlife
Article 51A(g): Fundamental duty to protect and improve natural environment

Major Acts

- Environment (Protection) Act, 1986 - Umbrella legislation
- Water (Prevention and Control of Pollution) Act, 1974
- Air (Prevention and Control of Pollution) Act, 1981
- Forest (Conservation) Act, 1980
- Wildlife Protection Act, 1972
- National Green Tribunal Act, 2010

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THE WATER (PREVENTION AND CONTROL OF POLLUTION) ACT, 1974

Objective of this Act:

- ❖ **Prevention and control of water pollution:** This involves implementing measures to reduce the discharge of pollutants into water bodies and to protect water quality from harmful contaminants that can affect ecosystems and human health.
- ❖ **Maintaining or restoring wholesomeness of water:** The Act aims to preserve the purity and usability of water for various purposes such as drinking, agriculture, industry, and recreation by preventing contamination and promoting treatment of polluted water.
- ❖ **Establishment of Boards for prevention and control of water pollution:** The Act provides for the creation of Central and State Pollution Control Boards. These Boards are responsible for monitoring water quality, enforcing pollution control standards, granting permissions for discharge of effluents, and taking necessary actions against violators to ensure compliance with the law.

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THE AIR (PREVENTION AND CONTROL OF POLLUTION) ACT, 1981

Objectives of this Act:

- ❖ **Prevention, Control, and Abatement of Air Pollution:** This act aims to prevent the introduction of pollutants into the atmosphere, control the emission of harmful substances, and reduce existing pollution levels to protect public health and the environment.
- ❖ **Maintaining the Quality of Air:** It establishes standards and guidelines to ensure that the air quality remains within safe limits, thereby safeguarding ecosystems and human well-being from the adverse effects of polluted air.
- ❖ **Establishment of Boards for the Prevention and Control of Air Pollution:** The act mandates the creation of central and state-level pollution control boards. These boards are responsible for monitoring air quality, enforcing regulations, conducting research, and promoting awareness about air pollution and its impacts.
- ❖ By addressing these objectives, the act plays a crucial role in India's environmental legislation framework, helping to mitigate the risks associated with air pollution and fostering sustainable development.

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THE ENVIRONMENT (PROTECTION) ACT, 1986

Objectives:

- ❖ **Protection and Improvement of Environment (Air, Water, Land):** This objective emphasizes safeguarding natural resources by controlling pollution and promoting sustainable practices. It involves monitoring and regulating pollutants to ensure air quality, water purity, and soil health are maintained or enhanced for current and future generations.
- ❖ **Prevention of Hazards to All Living Creatures (Humans, Animals, Plants):** The Act aims to minimize risks and harmful effects caused by environmental pollution or degradation. This includes preventing exposure to toxic substances, managing waste responsibly, and ensuring ecosystems remain balanced to support diverse life forms.
- ❖ **Maintenance of Harmonious Relationship Between Human Beings and Their Environment:** Recognizing the interdependence between humans and nature, this objective promotes sustainable development that meets human needs without compromising environmental integrity. It encourages practices that foster coexistence, respect for natural habitats, and responsible resource use to ensure long-term ecological balance.

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What is Environmental Law?

Environmental Law means the laws that regulate the impact of human activities on the environment. Environmental law covers a broad range of activities that affect air, water, land, flora, or fauna. It includes laws that relate to:

Protection of animals and plants	Agriculture and farming
Planning for the use and development of land	Waste management
Mining, exploration, and extractive industries	Climate change and emissions
Forestry	Weeds and invasive species
Pollution	Chemicals and pesticides
Fisheries	Conservation of natural and cultural heritage
Land and fire management	Water resource management (lakes, wetlands, rivers, and oceans)
Marine life	

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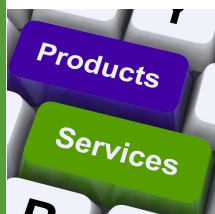
Targets of Environmental Laws: Who or what gets regulated?

1. Products:

The legislation targeted at products can be broad like the Toxic Substances Control Act (TSCA), which regulates any chemical substance or mixture. Or, legislation can be more specific like the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA), which only governs pesticide chemicals.

2. Pollutants:

Virtually every pollutant resulting from industrial, commercial, or domestic activities fall within the scope of a federal environmental statute. Industrial Facilities. Federal regulation targeted at industrial facilities is the most popular and perhaps the easiest form of regulation to enforce because facilities are fewer in number than individuals. For example, the EPA can more easily impose requirements for emissions reductions against car manufacturers rather than individual car owners.



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AIR POLLUTION



WATER POLLUTION



SOIL CONTAMINATION



NOISE POLLUTION

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Targets of Environmental Laws: Who or what gets regulated?

3. Government Agencies.

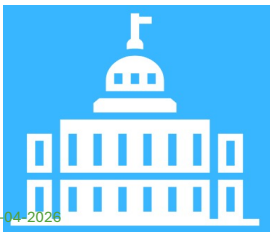
Governmental entities may be regulated if they own certain kinds of facilities that are part of the pollution problem, e.g. public water supply systems.

4. Individuals.

Regulation of individual conduct is less popular than regulation of facilities for both political and practical reasons.

5. Land Use:

While most private land use decisions are regulated only at the local level, some of the federal environmental laws contain provisions that effect land use decisions. The Clean Water Act, for example, requires individuals to obtain permits before dredge and fill operations are conducted on wetlands.



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Differences between Policy and Law



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Policy

It is a decision making frame work of course of action to achieve effect or change in the society

It is informal, as it is just a statement or a document of what it intended to be done in the future

Policies can lead to new law

Policies are made in the name of the people

They are administered by the state institutions or officials

Law

They are a set standards, principles, and procedures that must be followed in society

It is more formal as it is a system of rules and guidelines that are made for implementing justice

They can be modified/reviewed but not lead to new policies

Laws are made for the people

Laws are administered through the courts, which is enforceable in which the policies comply

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International Environmental Policy

Ways to help governments across the world to come to an agreement on environmental issues. They include:

1. United Nations
2. European Nations
3. World Trade Organization
4. World Bank
5. Non-governmental organization.



International environmental policy is supported by :

1. Customary law: laws/customs held by most cultures
2. Conventional Law: from conventions or treaties

Major International Treaties

- Antarctic Treaties System
- Convention on International Trade in Endangered Species of wildlife
- Kyoto Protocol
- Montreal on substance that deplete the ozone layer
- United Nations Framework Convention on climate change

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Some International Treaties

With growing recognition of the interconnections in our global environment, nations have become increasingly interested in signing on to international agreements (often called “treaties” or “conventions”) for environmental protection. A principal motivation in these treaties is the recognition that countries can no longer act alone to protect their own resources and interests.

CBD: Convention on Biological Diversity 1992 (1993)* **CITES:** Convention on International Trade on Endangered

CITES: Convention on International Trade on Endangered Species of Wild Fauna and Flora 1973 (1987)

CMS: Convention on the Conservation of Migratory Species of Wild Animals 1979 (1983)

Basel: Basel Convention on the Transboundary Movements of Hazardous Wastes and their Disposal 1989 (1992)

Ozone: Vienna Convention for the Protection of the Ozone Layer and Montreal Protocol on Substances that Deplete the Ozone Layer 1985 (1988)

UNFCCC: United Nations Framework Convention on Climate Change 1992 (1994)

CCD: United Nations Convention to Combat Desertification in those Countries Experiencing Serious Drought and/or Desertification, Particularly in Africa 1994 (1996)

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Environmental Protection Acts in India

With the rapid increase in industrialization and human needs, the environment has been badly suffered. Thankfully, the importance of protection, as well as conservation of the environment along with sustainable use of natural resources, is reflected really well in the constitutional framework of India. The Environmental laws in India give huge importance to maintaining an ecological balance of the environment by safeguarding the forests and wildlife of the country. There are certain constitutional provisions that give certain power and rights to the citizens to protect the environment.

- **Article 48A:** This Article comes under the Directive principle of the State policy. This article implies that State shall endeavor to protect the environment. It also emphasizes safeguarding the forests and wildlife of the country. Article 48A imposes a duty on State to protect the environment from pollution by adopting various measures.
- **Article 51A (g):** This Article states that it shall be the duty of each and every citizen of India to protect and improve the natural environment that includes lakes, rivers, forests, and wildlife. This Article also focuses on showing compassion for living creatures. This article is similar to Article 48A, but the only difference is that it concentrates on the fundamental duty of citizens whereas Article 48A instructs the state to perform its duties and protect the environment.
- **Article 253:** This Article gives power to Parliament to create laws for the country in order to implement any treaty conventions and agreements with other countries. By this article, Parliament enacted various laws in order to protect the environment like - Water Act 1974, Air Act 1981, and the Environmental Protection Act 1984.

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Environmental Protection Acts in India

- **Article 246:** This article divides the subjects of legislation between the Union and the States. It also provides details of the Concurrent list, in which both the Union and the States make laws, sharing jurisdiction over the protection of mines, wildlife, and mineral development. So, both State and Union have the power to enact laws to protect the environment. Article 246 also provides Parliament with extra power to make laws in the State list in the National interest.
- **Article 47:** This article imposes a duty on the State in order to improve the standards of living of citizens by providing health facilities, proper nutrition, and sanitization and protecting the environment to live safely. Article 47 also urges its citizens to be more environmentally conscious.
- **Article 21:** It states that the right to life is not just for animals but also provides the right to humans to live safely in an environment with basic human dignity. Because. In M.C. Mehta vs. Union of India, the Supreme Court had stated that the right to live includes living in a pollution-free environment and being free from diseases.
- **Article 19(1) (g):** It states that citizens cannot practice such trade or business activities that are hazardous to public health.
- **Article 32 & 226:** This article provides the right to citizens to approach to Supreme or High Court whenever there is a violation of fundamental rights by PIL (Public Interest Litigation). This article helps preserve the environment and maintain ecological balance. This Article also dictates that environmental conservation is not just the duty of the government but also the responsibility of the citizens of India.

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Major Environment Policies and Legislations in India and their Goals

The Wildlife Protection Act, 1972



- This act introduced rational and modern wildlife management
- It has six schedules that give varying degree of protection
- Statutory bodies under WPA:
 - National Board for wildlife and state wildlife advisory boards
 - Central zoo authority
 - Wildlife Crime Control Bureau
 - National Tiger Authority

State wise Bengal tiger population India as of 2018.



Water Conservation and Control of Pollution Act, 1974

- To provide prevention and control of water pollution.
- Maintaining or restoring of wholesomeness and purity of water in various sources of water.
- Statutory bodies:
 - Centre Pollution Control Boards (CPCB), and
 - State Pollution Control Board (SPCB).
- CPCB and SPCB establish and enforce effluent standards for factories discharging pollutants into water bodies.

Taxation

Water (Prevention and Control of Pollution) Cess Act, 1977

Water (Prevention and Control of Pollution) Cess Rules, 1978

Groundwater

Model Bill to Regulate and Control the Development and Management of Groundwater

Effluent standards

Environmental (Protection) Act, 1986

Environmental (Protection) Rules, 1986

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Major Environment Policies and Legislations in India and their Goals

The Forest Conservation Act, 1980



- It extends to India except the State of Jammu and Kashmir.
- Came into force on the 25th day of October 1980.
- The Parliament of India ensures the conservation of forests and its resources
- The aim of the Act is to preserve the forest ecosystem of India by fulfilling the following objectives:
 - Protect the forest along with its flora, fauna and other diverse ecological components while preserving the integrity and territory of the forests.
 - Arrest the loss of forest biodiversity
 - Prevent forest lands being converted into agricultural, grazing or for any other commercial purposes and intentions.

Air Prevention and Control of Pollution act, 1981

- To provide for prevention, control, and abatement of air pollution
- To provide for the establishment of the boards at the central and state levels to implement the act.
- CPCB and SPCB were given the responsibility.

The Environment Protection Act, 1986

- It focuses on the conservation of the environment and the purpose of plugging the rules in the existing legislation.
- This act was passed under article 253 (legislation for giving effect to international agreements)



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Major Environment Policies and Legislations in India and their Goals

The Hazardous Waste (Management and Handling), 1989

- To control the generation, collection, treatment, import, storage, and handling of hazardous waste.
- These Rules came into effect in the year 1989
- The Rules lay down corresponding duties of various authorities such as MoEF, CPCB, State/UT Govts., SPCBs/PCCs, DGFT, Port Authority and Custom Authority while State Pollution Control Boards/ Pollution Control Committees



Public Liability Insurance Act, 1991

- provides compulsory insurance to provide immediate relief to the people who are affected by accidents while handling any dangerous substance.

The Biomedical waste (Management and Handling) Rules, 1998



- to streamline the process of proper handling of hospital waste
- disposal, collection, and treatment.
- These rules apply to all persons who generate, collect, receive, transport, treat, dispose, or handle bio-medical waste in any form
- The Government of every State/Union Territory shall constitute an Advisory Committee. The Committee will include experts in the fields of
- medical and health, animal husbandry and veterinary sciences, environmental management, municipal administration, and any other related department or organization including non-governmental organizations



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Major Environment Policies and Legislations in India and their Goals

The Energy Conservation Act, 2001



- It was enacted as a step towards improving energy efficiency and reducing wastage. It specifies the energy consumption standards for equipment and appliances.
- It prescribes energy consumptions norms and standards for consumers. It prescribes energy conservation building codes for commercial buildings.
- statutory body:
 - Bureau of energy efficiency (BEE)

The Noise Pollution (Regulation and Control) Amendment, 2002

- Rules lay down such terms and conditions as are necessary to regulate noise pollution, permit the use of loudspeakers or public address systems during night hours on or during any cultural or religious festival



The Biological Diversity Act, 2002



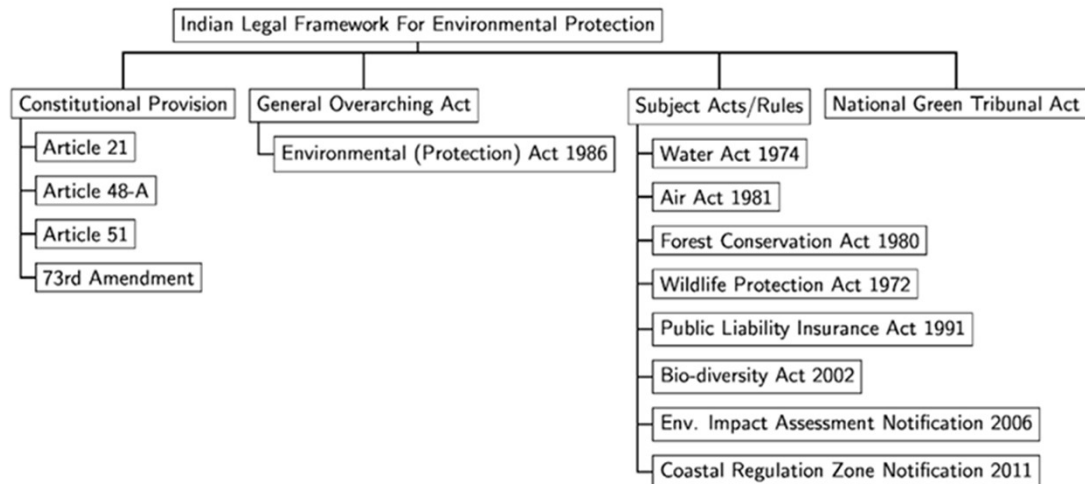
- To realize the objectives enshrined in the United Nations Convention on Biological Diversity (CBD), 1992 which recognizes the sovereign rights of states to use their own Biological Resources.
- The Act aims at the conservation of biological resources and associated knowledge as well as facilitating access to them in a sustainable manner
- It was implemented to give effect to CBD, Nagoya Protocol.
 - To set up National Biodiversity Authority (NBA), State Biodiversity Boards (SBBS), and Biodiversity Management Committees (BMCS).

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Environmental Protection Act In India



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Approaches to Environmental Policy

1. Tax Breaks & Subsidies

Tax breaks for industries that perform environmentally-friendly actions (reducing emission, solar panels)

2. Green Taxes:

Taxes are levied on companies for their harmful environmental impacts.

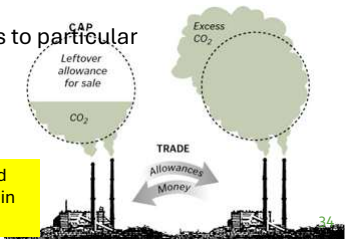
Money that is collected is used to clean up the area

Attempts to encourage companies to change their bad practices

3. Cap-and-Trade

The government decides the maximum quantities of a pollutant and issues permits to particular companies

Can be bought, sold, and traded



The cap-and-trade approach in the U.S. has helped reduce sulfur dioxide emissions that cause acid rain by 35%

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Approaches to Environmental Policy

4. Local Incentives

Efforts on the local level to charge for waste disposal and rebates for water-efficient toilets, Energy Star appliances, etc.

5. Command-and-control:

Environmental policy sets rules or limits and threatens punishment for violators.

It relies on standards of various types to improve environmental quality.

Most environmental laws have resulted in safe, healthy, and comfortable lives.



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Step 1
Get the right
people involved



Step 2
Think the program
through



Step 3
Implement the
program

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Carbon Credit Trading Scheme (CCTS)

India is advancing its climate policy by implementing the Carbon Credit Trading Scheme (CCTS) and the Green Credit Program, aiming to reduce greenhouse gas emissions and promote sustainable practices.

Regulatory Framework: Established under the Energy Conservation (Amendment) Act, 2022, the CCTS empowers the government to create a domestic carbon market. The Ministry of Power oversees the scheme, with the Bureau of Energy Efficiency (BEE) serving as the designated administrator.

1. Green Credit Program:

Launch and Objectives: Introduced in October 2023 as part of the Lifestyle for Environment (LiFE) initiative, this market-based mechanism incentivizes voluntary environmental actions across various sectors, including individuals, societies, industries, and private companies.

Scope of Activities: The program encompasses eight key areas: tree plantation, water conservation, sustainable agriculture, waste management, air pollution reduction, mangrove conservation, eco-friendly product certification (Eco Mark), and sustainable building and infrastructure development.

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Why is enforcement of Environmental Laws Difficult?

Environmental Policies are challenged, ignored, and rejected by citizens because:

1. Poor coordination across government agencies.
2. Weak institutional capacity.
3. Lack of access to information.
4. Environmental Policy involves the government regulation of property owners and businesses; people think regulations are inconvenient and cause economic loss.
5. Corruption and stifled civic engagement are the key factors behind the poor effectiveness and implementation of environmental regulations
6. Problems develop gradually, over the long term, but human behavior is geared to short-term needs
 1. Businesses opt for short-term economic gain
 2. News media have a short attention span
 3. Politicians act out of short-term interest.



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CASE STUDIES

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Case Study: London Congestion Charge

World's largest urban road pricing scheme • Launched February 2003

Objectives

- Reduce traffic congestion
- Improve air quality
- Fund public transport

Results

- 30% reduction in traffic
- 20% decrease in CO₂ emissions
- £1.2 billion revenue for transport

Lessons Learned

- Strong political commitment essential
- Public transport alternatives must be available
- Technology and enforcement systems critical
- Revenue reinvestment builds public support

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Case Study: Costa Rica Forest Conservation

From 21% forest cover (1987) to 53% (2020) - A conservation success story

53%

Forest cover (2020)

\$500M

PES payments since 1997

6%

Of global biodiversity

Key Policy Instruments

- Payment for Ecosystem Services (PES) - Pays landowners for conservation
- Ban on deforestation (1996)
- Protected areas expanded to 25% of territory
- Ecotourism development - Now second-largest revenue source

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Case Study: California Climate Leadership

Subnational climate action model - 5th largest economy implementing ambitious policies

AB 32 (2006)

Global Warming Solutions Act - Cap emissions at 1990 levels by 2020

SB 100 (2018)

100% clean electricity by 2045

Cap-and-Trade

Launched 2013, covers 80% of state emissions

ZEV Mandate

100% new car sales zero-emission by 2035

Building Standards

Net-zero energy for new homes by 2020

Impact: 25% emission reduction since 2006 while economy grew 78%

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Case Study: China's War on Pollution

Aggressive action since 2013 to tackle severe air pollution crisis

Major Initiatives

- National Action Plan on Air Pollution (2013) - Targets for PM2.5 reduction
- Coal consumption cap - Reduced by 400 million tonnes 2013-2017
- Industrial upgrades - Closed thousands of polluting factories
- Vehicle emission standards - China 6 equivalent to Euro 6
- Renewable energy investment - World's largest solar and wind capacity
- Environmental Protection Law (2015) - Stricter enforcement and penalties

Results: 40% reduction in PM2.5 levels in major cities (2013-2020)

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Case Study: Rwanda's Plastic Bag Ban

The Policy (2008)

- Complete ban on plastic bags
- Airport inspections and confiscation
- Fines and imprisonment for violations
- Monthly community cleanup days

Impact

- One of Africa's cleanest countries
- Shift to reusable bags
- Tourism boost
- Model for other African nations

Success Factors

- Strong government commitment and enforcement
- Public awareness campaigns
- Alternative solutions promoted (paper bags, baskets)
- Integration with broader environmental vision

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Over 60 countries have now implemented similar bans

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Case Study: Flint Water Crisis

Environmental justice and regulatory failure - Michigan, USA (2014-2016)

Crisis Begins	April 2014: Water source switched to Flint River without corrosion control
Problems Emerge	Residents report discolored water, rashes, hair loss; officials dismiss concerns
Lead Discovered	Sept 2015: Pediatrician finds elevated lead levels in children's blood
Emergency Declared	Oct 2015: State of emergency; return to Detroit water supply
Response	2016-2021: Lead pipe replacement program; bottled water distribution

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Flint Crisis: Lessons Learned

Regulatory Failures

EPA oversight inadequate • State environmental agency dismissed warnings • Lack of corrosion control requirements enforcement

Environmental Justice

Poor, minority community disproportionately affected • Concerns initially ignored • Economic considerations prioritized over health

Public Health

12,000+ children exposed to lead • Irreversible neurological damage • Legionnaires' disease outbreak (12 deaths)

Policy Changes

Federal Lead and Copper Rule revised • Michigan lead testing protocols strengthened • Water infrastructure funding increased

Key Takeaways

Listen to affected communities • Preventive measures cost less than remediation • Transparency essential • Environmental justice matters

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Environmental Justice

Fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income with respect to environmental laws and policies

Disparities

- Low-income communities bear greater pollution burden
- Minority neighborhoods more likely near hazardous sites
- Less access to clean air, water, green space
- Higher rates of asthma, cancer, other diseases

Policy Responses

- Executive Order 12898 (US, 1994)
- Environmental justice screening tools
- Community participation requirements
- Cumulative impact assessments

Recent Developments

Biden Administration (2021): Justice40 Initiative - 40% of benefits from climate and clean energy investments to disadvantaged communities

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Enforcement Mechanisms

Administrative Orders	Compliance orders, penalties without court action
Civil Penalties	Fines for violations, typically per day; can reach millions
Criminal Prosecution	For knowing violations, false statements; prison terms possible
Injunctive Relief	Court orders to stop violations or take corrective action
Permit Suspension/Revocation	Remove authorization to operate
Citizen Suits	Private enforcement when government fails to act
Supplemental Environmental Projects	Environmentally beneficial projects in lieu of penalties

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Compliance Monitoring Approaches

Self-Reporting

Companies submit monitoring data • Discharge monitoring reports • Emissions inventories

Inspections

On-site facility visits • Record reviews • Sampling and testing

Continuous Monitoring

Real-time emissions tracking • Automatic data transmission • Early violation detection

Remote Sensing

Satellite imagery • Drone surveillance • Leak detection technology

Community Monitoring

Citizen science • Air quality sensors • Complaint systems

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Environmental Impact Assessment (EIA)

Systematic process to identify, predict, and evaluate environmental effects of proposed projects

- 1 Screening - Determine if EIA required
- 2 Scoping - Identify key issues and impacts
- 3 Impact Analysis - Assess magnitude and significance
- 4 Mitigation - Propose measures to reduce impacts
- 5 Reporting - Document findings in EIS/EA
- 6 Review - Public and agency comment period
- 7 Decision - Approval, conditions, or rejection
- 8 Monitoring - Track actual impacts post-approval

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FUTURE DIRECTIONS & CHALLENGES

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Emerging Environmental Challenges

Microplastics	Ubiquitous pollution • Health impacts unknown • No effective regulation yet
PFAS Chemicals	'Forever chemicals' in water • Widespread contamination • Emerging standards
E-Waste	Fastest growing waste stream • Toxic materials • Inadequate recycling
Biodiversity Loss	6th mass extinction • Ecosystem collapse • Limited legal protection
Ocean Acidification	CO ₂ absorption • Coral reef destruction • Marine life impacts
Space Debris	Growing problem • No international framework • Collision risks

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Technology & Innovation in Environmental Policy

Artificial Intelligence	Emissions monitoring • Waste sorting • Climate modeling • Biodiversity tracking
Blockchain	Carbon credit verification • Supply chain transparency • Renewable energy trading
IoT Sensors	Real-time pollution monitoring • Water quality tracking • Smart grids
Satellite Technology	Deforestation detection • Ocean monitoring • Illegal dumping surveillance
Green Chemistry	Safer alternatives • Biodegradable materials • Pollution prevention
Carbon Capture	Direct air capture • Point source capture • Storage technology

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International Cooperation Challenges

Sovereignty Concerns

Nations reluctant to cede control over domestic policies

Economic Disparities

Developed vs developing country responsibilities and capacities

Free Rider Problem

Benefits of environmental protection are global; costs are local

Enforcement Gaps

Limited mechanisms to compel compliance with international agreements

Political Instability

Changes in government can undermine long-term commitments

Competing Priorities

Economic growth, security, health may take precedence

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Integrated Environmental Policy

Moving beyond siloed approaches to address interconnected challenges

Climate-Water Nexus

Droughts • Floods • Hydropower • Agricultural water use

Energy-Environment

Renewable transition • Storage • Grid modernization • Land use

Food-Land-Biodiversity

Sustainable agriculture • Habitat preservation • Soil health

Circular Economy

Waste → resource • Product design • Supply chains

Health-Environment

Air quality • Water safety • Chemical exposure • Climate impacts

Urban Planning

Green spaces • Public transport • Building efficiency • Resilience

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Green Finance & Investment

\$5.2 trillion

Global ESG assets (2022)

\$540 billion

Green bond issuance (2022)

\$100 billion

Climate finance commitment

Financing Mechanisms

- Green bonds - Fixed income for environmental projects
- Sustainability-linked loans - Interest rates tied to ESG targets
- Carbon markets - Trading emission allowances and offsets
- Climate funds - Public financing for developing countries
- ESG investing - Environmental, social, governance criteria
- Blended finance - Public-private partnership structures

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Corporate Environmental Responsibility

Voluntary Initiatives

- Science Based Targets initiative
- RE100 (100% renewable energy)
- B Corp certification
- UN Global Compact
- CDP disclosure

Mandatory Disclosure

- EU Taxonomy Regulation
- TCFD recommendations
- SEC climate disclosure rules
- UK Streamlined Energy & Carbon Reporting
- Supply chain due diligence laws

Trends

- Shift from voluntary to mandatory requirements
- Standardization of ESG metrics and reporting
- Greenwashing scrutiny and enforcement
- Stakeholder capitalism replacing shareholder primacy

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Public Participation in Environmental Policy

Aarhus Convention Pillars (1998)

Access to Information

Right to environmental data and documents from public authorities

Public Participation

Involvement in decisions on specific activities and plans/policies

Access to Justice

Right to challenge decisions and seek remedies in courts

Participation Methods

Public hearings • Comment periods • Advisory committees • Citizen science • Environmental NGOs • Social media engagement

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Evaluating Policy Effectiveness

Environmental Outcomes	Emissions reduction • Species recovery • Water quality improvement • Habitat protection
Compliance Rates	Permit violations • Enforcement actions • Self-reporting accuracy
Cost-Effectiveness	\$/tonne CO ₂ reduced • Cost per life saved • Economic efficiency
Implementation	Timeline adherence • Stakeholder engagement • Institutional capacity
Innovation	Technology development • Market transformation • Behavioral change
Equity	Distributional impacts • Environmental justice • Access to benefits

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Environmental Policy Instruments

Selecting the right tool for the environmental challenge

Command-and-Control

Technology standards, emission limits

+ Clear requirements

- Less flexible

Market-Based

Taxes, cap-and-trade, subsidies

+ Cost-effective, innovation

- Complex design

Information

Labeling, disclosure, education

+ Low cost, empowers choice

- Slow behavior change

Voluntary

Industry agreements, eco-labels

+ Stakeholder buy-in

- Enforcement challenges

Property Rights

Tradable permits, fishing quotas

+ Efficiency, flexibility

- Initial allocation issues

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Adaptive Environmental Management

Iterative approach to policy in face of uncertainty and changing conditions

Assess

Gather data on environmental conditions and policy impacts

Design

Develop policy based on best available science

Implement

Execute policy with monitoring mechanisms

Monitor

Track outcomes and emerging issues

Evaluate

Compare results to objectives

Adjust

Modify policy based on learning

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Environmental SDGs

UN Sustainable Development Goals with environmental components

SD
G 6

Clean Water & Sanitation

Universal access by 2030

SD
G 7

Affordable & Clean Energy

Renewable energy increase

SD
G 12

Responsible Consumption

Circular economy principles

SD
G 13

Climate Action

Limit warming, build resilience

SD
G 14

Life Below Water

Conserve oceans, marine resources

SD
G 15

Life on Land

Protect ecosystems, halt biodiversity loss

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Nature-Based Solutions

Using natural processes to address environmental and societal challenges

Reforestation	Carbon sequestration • Biodiversity • Water regulation • Livelihoods
Wetland Restoration	Flood control • Water filtration • Carbon storage • Habitat
Urban Green Infrastructure	Heat reduction • Air quality • Stormwater management • Wellbeing
Mangrove Protection	Coastal protection • Carbon sink • Fisheries • Tourism
Agroforestry	Soil health • Crop resilience • Income diversification • Carbon
Green Roofs/Walls	Building insulation • Air quality • Urban biodiversity • Aesthetics

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Policy Recommendations

- Strengthen international cooperation and enforcement mechanisms
- Integrate environmental considerations across all policy sectors
- Price carbon and other environmental externalities
- Invest in green infrastructure and nature-based solutions
- Accelerate renewable energy transition with supporting policies
- Mandate corporate environmental disclosure and accountability
- Ensure environmental justice in policy design and implementation
- Build adaptive capacity to respond to climate impacts
- Support research, innovation, and technology transfer
- Engage communities and stakeholders in decision-making
- Increase funding for environmental protection and restoration
- Harmonize environmental standards globally while respecting local contexts

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Barriers to Policy Implementation

Political	Short-term election cycles • Lobbying • Ideology • Partisan divides
Economic	Upfront costs • Competitive disadvantages • Job losses • Budget constraints
Institutional	Fragmented authority • Lack of capacity • Bureaucratic inertia
Technical	Knowledge gaps • Technology limitations • Data availability
Social	Public opposition • Behavioral resistance • Equity concerns • Cultural factors
Legal	Jurisdictional conflicts • Property rights • Enforcement challenges

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Factors for Successful Environmental Policy

Political Will	Strong leadership • Cross-party support • Long-term commitment
Scientific Basis	Evidence-based • Expert input • Adaptive to new knowledge
Clear Objectives	Measurable targets • Timelines • Accountability
Stakeholder Buy-in	Consultation • Transparency • Shared benefits
Adequate Resources	Funding • Personnel • Infrastructure • Technology
Effective Enforcement	Monitoring • Penalties • Compliance incentives
Public Support	Education • Communication • Visible benefits

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The Road to 2030

Critical decade for environmental action

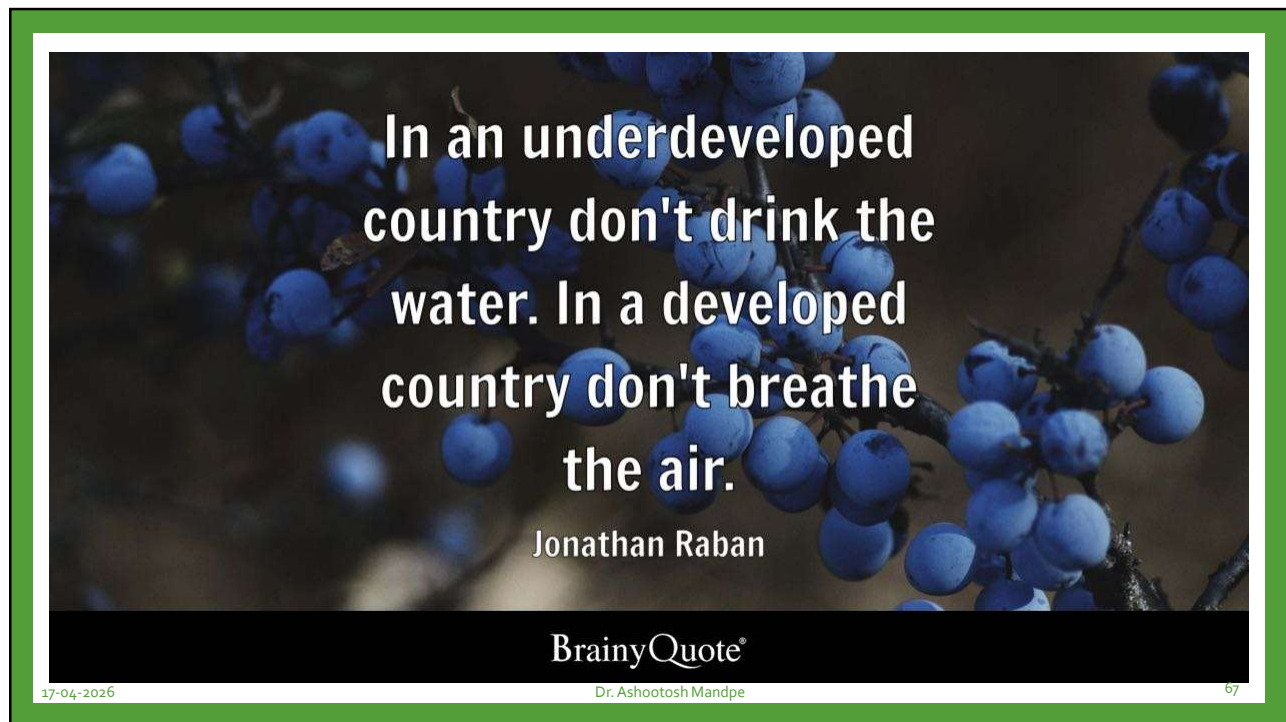
Climate	Halve global emissions • Scale renewables to 60%+ • Phase out coal
Biodiversity	Protect 30% land and sea • Halt species loss • Restore ecosystems
Pollution	Eliminate single-use plastics • Clean air in all cities • Zero toxic releases
Circular Economy	50% waste reduction • Design out waste • Close material loops
Finance	Align all finance flows • End fossil fuel subsidies • \$trillions to green economy

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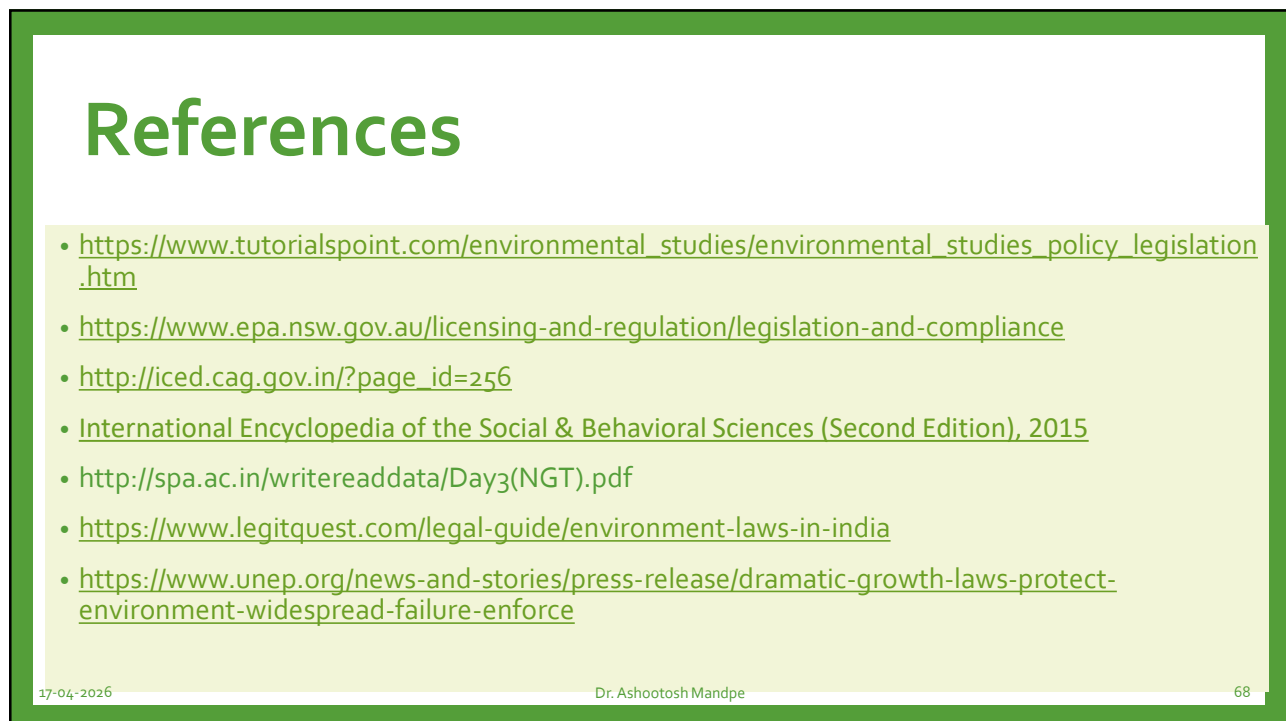
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Thank You!
धन्यवाद

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